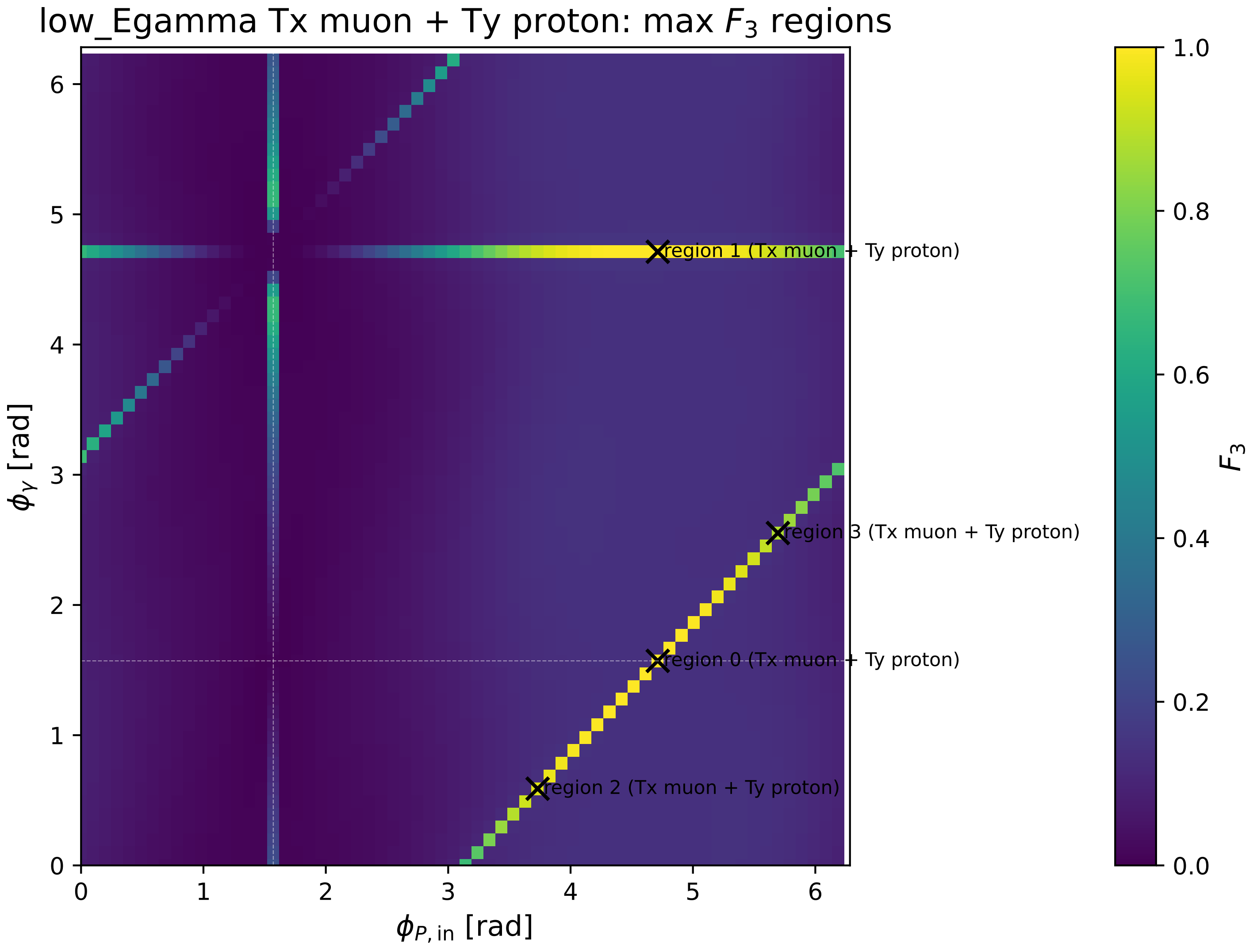
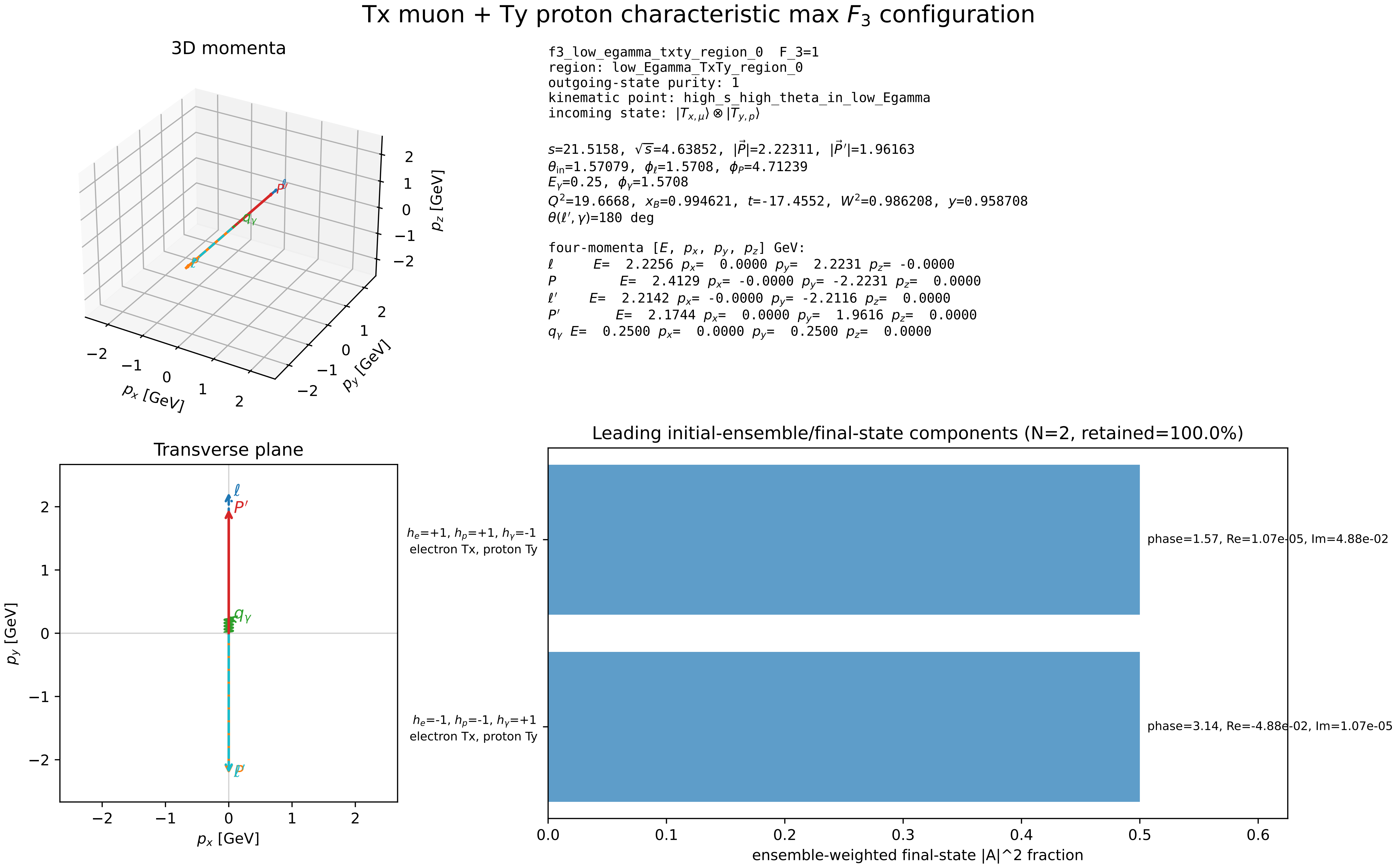
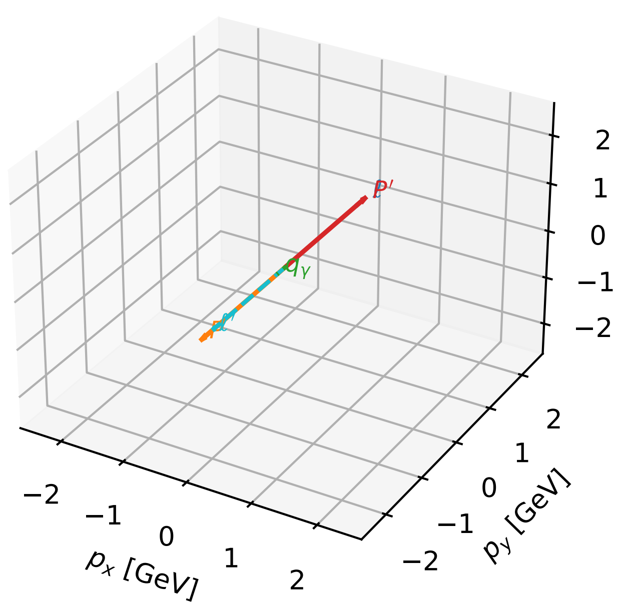

low_Egamma Tx muon + Ty proton: max $F_3$ regions
$\phi_\gamma$ [rad]
$\phi_{P,\mathrm{in}}$ [rad]
region 1 (Tx muon + Ty proton)
region 3 (Tx muon + Ty proton)
region 0 (Tx muon + Ty proton)
region 2 (Tx muon + Ty proton)
$F_3$
0
1
2
3
4
5
6
0.0
0.2
0.4
0.6
0.8
1.0


Tx muon + Ty proton characteristic max $F_3$ configuration
3D momenta
$p_x$ [GeV]
$p_y$ [GeV]
$p_z$ [GeV]
$P'$
$\ell$
$q_\gamma$
$\ell'$
f3_low_egamma_txty_region_0  F_3=1
region: low_Egamma_TxTy_region_0
outgoing-state purity: 1
kinematic point: high_s_high_theta_in_low_Egamma
incoming state: $|T_{x,\mu}\rangle \otimes |T_{y,p}\rangle$
$s$=21.5158, $\sqrt{s}$=4.63852, $|\vec{P}|$=2.22311, $|\vec{P}'|$=1.96163
$\theta_{\rm in}$=1.57079, $\phi_\ell$=1.5708, $\phi_P$=4.71239
$E_\gamma$=0.25, $\phi_\gamma$=1.5708
$Q^2$=19.6668, $x_B$=0.994621, $t$=-17.4552, $W^2$=0.986208, $y$=0.958708
$\theta(\ell', \gamma)$=180 deg
four-momenta [$E$, $p_x$, $p_y$, $p_z$] GeV:
$\ell$ $E$= 2.2256 $p_x$= 0.0000 $p_y$= 2.2231 $p_z$= -0.0000
$P$ $E$= 2.4129 $p_x$= -0.0000 $p_y$= -2.2231 $p_z$= 0.0000
$\ell'$ $E$= 2.2142 $p_x$= -0.0000 $p_y$= -2.2116 $p_z$= 0.0000
$P'$ $E$= 2.1744 $p_x$= 0.0000 $p_y$= 1.9616 $p_z$= 0.0000
$q_\gamma$ $E$= 0.2500 $p_x$= 0.0000 $p_y$= 0.2500 $p_z$= 0.0000
Transverse plane
$p_x$ [GeV]
$p_y$ [GeV]
Leading initial-ensemble/final-state components (N=2, retained=100.0%)
$h_e$=+1, $h_p$=+1, $h_\gamma$=-1
electron Tx, proton Ty
phase=1.57, Re=1.07e-05, Im=4.88e-02
$h_e$=-1, $h_p$=-1, $h_\gamma$=+1
electron Tx, proton Ty
phase=3.14, Re=-4.88e-02, Im=1.07e-05
ensemble-weighted final-state |A|^2 fraction

# Tx muon + Ty proton characteristic max $F_3$ configuration

## 3D momenta

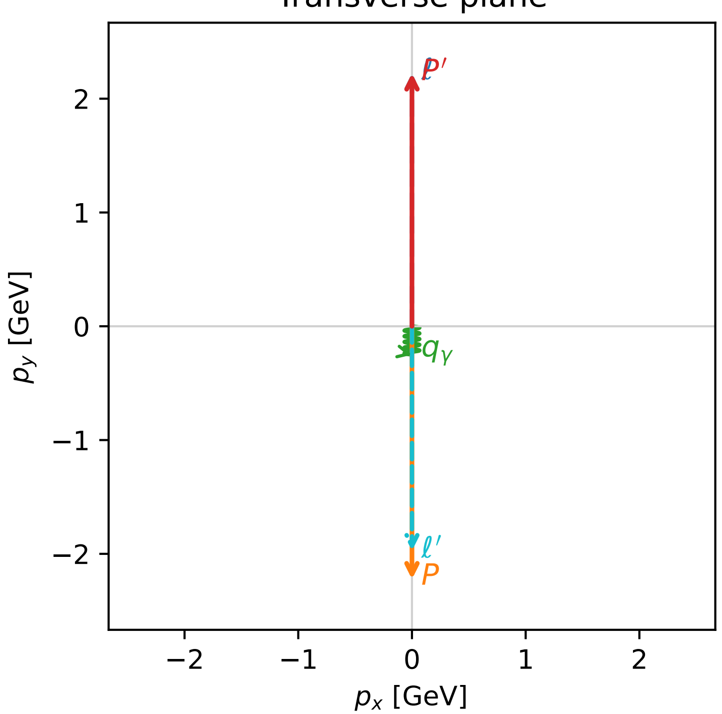



f3_low_egamma_txty_region_1  F_3=0.999999
region: low_Egamma_TxTy_region_1
outgoing-state purity: 1
kinematic point: high_s_high_theta_in_low_Egamma
incoming state: $|T_{x,\mu}\rangle \otimes |T_{y,p}\rangle$

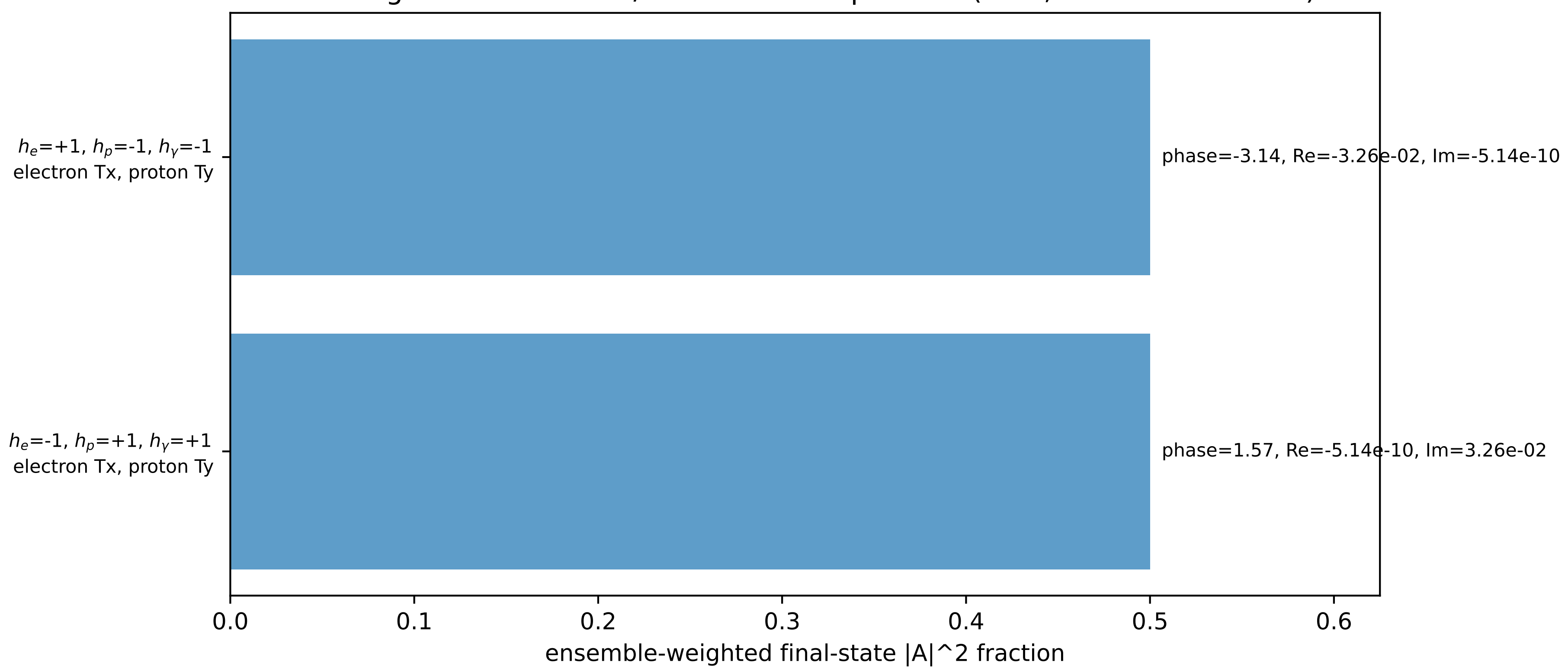
$s$=21.5158, $\sqrt{s}$=4.63852, $|\vec{P}|$=2.22311, $|\vec{P}'|$=2.22295
$\theta_{\text{in}}$=1.57079, $\phi_\ell$=1.5708, $\phi_P$=4.71239
$E_\gamma$=0.25, $\phi_\gamma$=4.71239
$Q^2$=17.5445, $x_B$=0.883304, $t$=-19.7674, $W^2$=3.19769, $y$=0.963029
$\theta(\ell', \gamma)$=0 deg

four-momenta [$E$, $p_x$, $p_y$, $p_z$] GeV:
$\ell$ $E$= 2.2256 $p_x$= 0.0000 $p_y$= 2.2231 $p_z$= -0.0000
$P$ $E$= 2.4129 $p_x$= -0.0000 $p_y$= -2.2231 $p_z$= 0.0000
$\ell'$ $E$= 1.9758 $p_x$= 0.0000 $p_y$= -1.9729 $p_z$= 0.0000
$P'$ $E$= 2.4127 $p_x$= 0.0000 $p_y$= 2.2229 $p_z$= 0.0000
$q_\gamma$ $E$= 0.2500 $p_x$= -0.0000 $p_y$= -0.2500 $p_z$= 0.0000

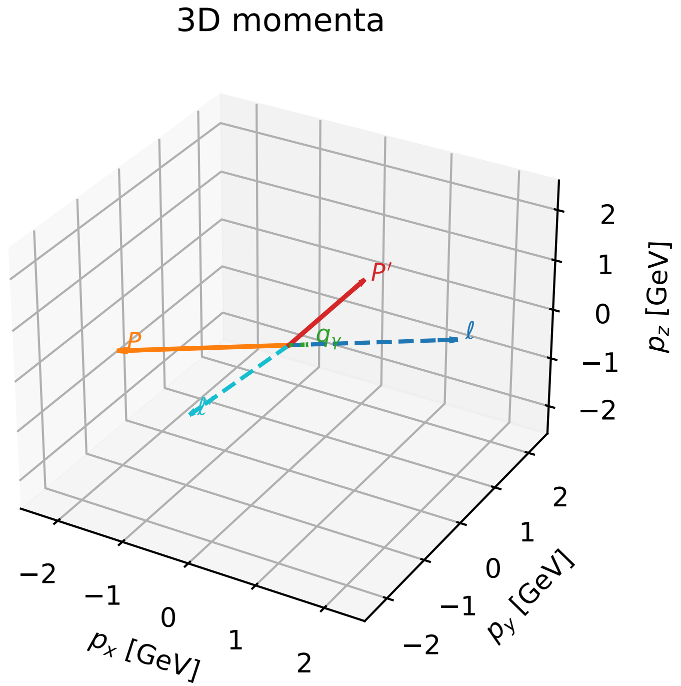
## Transverse plane




## Leading initial-ensemble/final-state components (N=2, retained=100.0%)



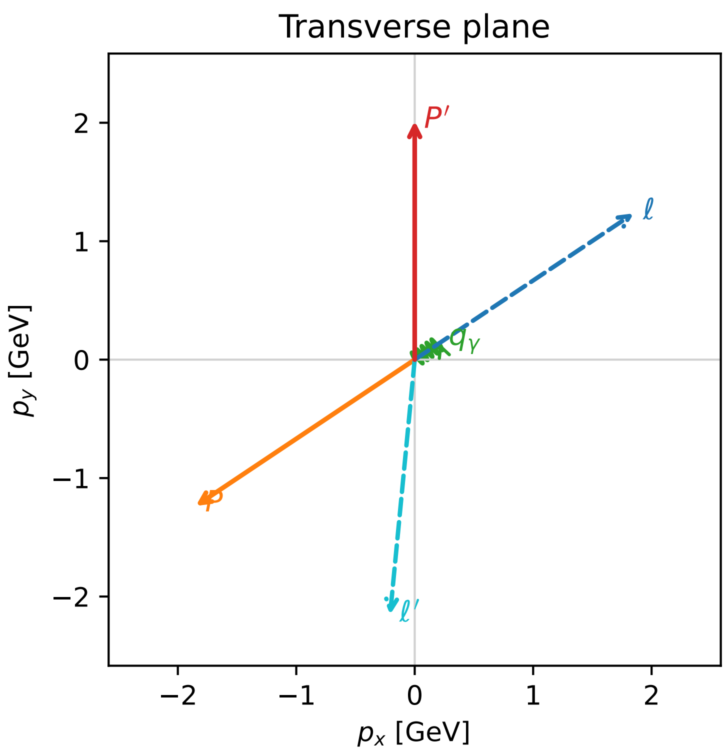
# Tx muon + Ty proton characteristic max $F_3$ configuration

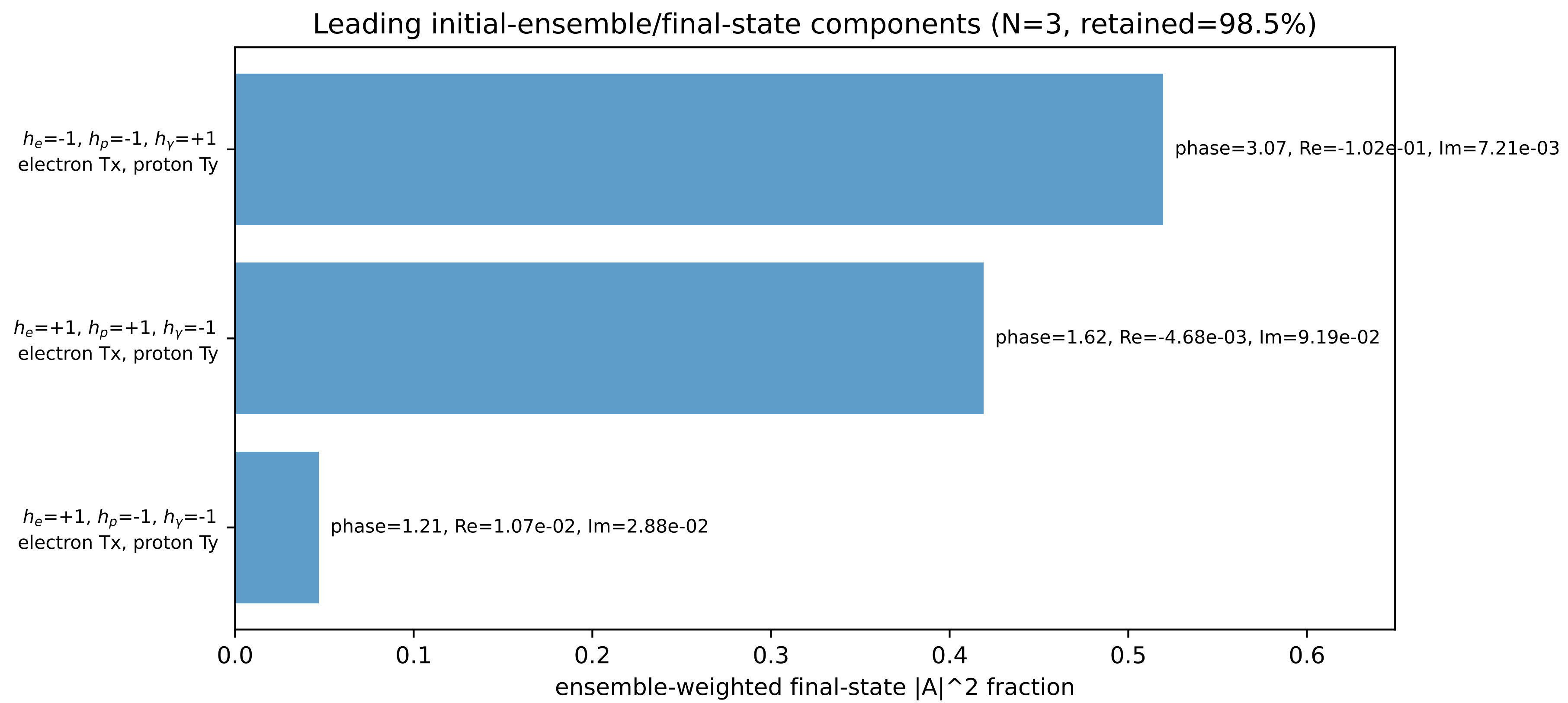



```
f3_low_egamma_txty_region_2  F_3=0.948086
region: low_Egamma_TxTy_region_2
outgoing-state purity: 1
kinematic point: high_s_high_theta_in_low_Egamma
```
incoming state: $|T_{x,\mu}\rangle \otimes |T_{y,p}\rangle$

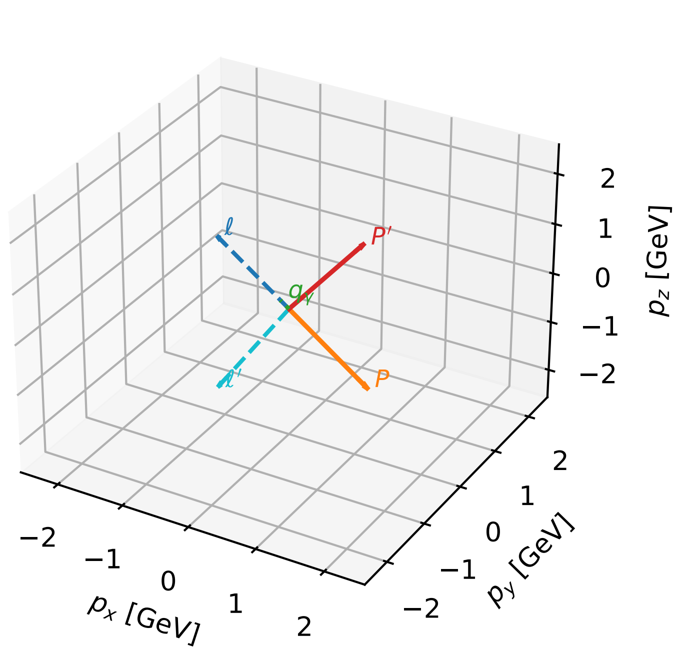
$s$=21.5158, $\sqrt{s}$=4.63852, $|\vec{P}|$=2.22311, $|\vec{P}'|$=2.01469
$\theta_{\mathrm{in}}$=1.57079, $\phi_\ell$=0.589049, $\phi_P$=3.73064
$E_\gamma$=0.25, $\phi_\gamma$=0.589049
$Q^2$=15.708, $x_B$=0.96608, $t$=-13.9416, $W^2$=1.43137, $y$=0.788349
$\theta(\ell',\gamma)$=129.263 deg

four-momenta [$E$, $p_x$, $p_y$, $p_z$] GeV:
$\ell$ $E$= 2.2256 $p_x$= 1.8484 $p_y$= 1.2351 $p_z$= -0.0000
$P$ $E$= 2.4129 $p_x$= -1.8484 $p_y$= -1.2351 $p_z$= 0.0000
$\ell'$ $E$= 2.1662 $p_x$= -0.2079 $p_y$= -2.1536 $p_z$= 0.0000
$P'$ $E$= 2.2223 $p_x$= 0.0000 $p_y$= 2.0147 $p_z$= 0.0000
$q_\gamma$ $E$= 0.2500 $p_x$= 0.2079 $p_y$= 0.1389 $p_z$= 0.0000



# Tx muon + Ty proton characteristic max $F_3$ configuration




f3_low_egamma_txty_region_3  F_3=0.881818
region: low_Egamma_TxTy_region_3
outgoing-state purity: 1
kinematic point: high_s_high_theta_in_low_Egamma
incoming state: $|T_{x,\mu}\rangle \otimes |T_{y,p}\rangle$

$s$=21.5158, $\sqrt{s}$=4.63852, $|\vec{P}|$=2.22311, $|\vec{P}'|$=2.01469
$\theta_{\mathrm{in}}$=1.57079, $\phi_\ell$=2.55254, $\phi_P$=5.69414
$E_\gamma$=0.25, $\phi_\gamma$=2.55254
$Q^2$=15.708, $x_B$=0.96608, $t$=-13.9416, $W^2$=1.43137, $y$=0.788349
$\theta(\ell', \gamma)$=129.263 deg

four-momenta [$E$, $p_x$, $p_y$, $p_z$] GeV:
$\ell$ $E$= 2.2256 $p_x$= -1.8484 $p_y$= 1.2351 $p_z$= -0.0000
$P$ $E$= 2.4129 $p_x$= 1.8484 $p_y$= -1.2351 $p_z$= 0.0000
$\ell'$ $E$= 2.1662 $p_x$= 0.2079 $p_y$= -2.1536 $p_z$= 0.0000
$P'$ $E$= 2.2223 $p_x$= 0.0000 $p_y$= 2.0147 $p_z$= 0.0000
$q_\gamma$ $E$= 0.2500 $p_x$= -0.2079 $p_y$= 0.1389 $p_z$= 0.0000